1(a)	Impulse $= \Delta p$ $= p_{final} - p_{initial}$ $= -3.4 - 6.0$ $= \underline{-9.4 \text{ kg m s}^{-1}}$ (2 sf)	 **M1** **A1**
(b)	$F = \Delta p / \Delta t$ $= -9.4 / 0.2$ $= -47$ N acceleration $= -47 / 0.30$ $= -156.67$ $= \underline{-160 \text{ m s}^{-2}}$ (2 sf / accept positive answer here) direction : opposite	 **C1** **A1** **A1**
(c)	Initial KE $= p_{initial}^2 / 2m$ $= 6.0^2 / (2 \times 0.30)$ $= 60$ J Final KE $= p_{final}^2 / 2m$ $= 3.4^2 / (2 \times 0.30)$ $= 19.267$ J Change in KE $= 60 - 19.267 = 40.733$ J $= \underline{41 \text{ J}}$ (0 dp)	 **C1** **A1**
(d)	The collision is **inelastic** because the **kinetic energy of the ball decreases** after collision. Can accept any of the following too: • speed ball decreases • momentum of ball decreases in magnitude • relative velocities of separation is smaller than the relative velocities of approach • energy of ball is not conserved	**B1**